Queue Problem Bank

```
Legend
★ = Core Placement Problem (Mandatory)
Language-Agnostic DSA Problem Bank | Implement all Queue variants from scratch before
using built-in structures.
```

Section 1: Queue Internals — Implementation

★ **Implement Linear Queue using Array (Enqueue, Dequeue, Front, Rear, isEmpty, isFull)**
★ **Implement Queue using Linked List**
★ **Implement Circular Queue using Array**
★ **Implement Deque (Double-Ended Queue) — Insert and Delete from Both Ends**
Implement Priority Queue using Array (Min and Max)
★ **Display All Elements of a Queue (Traverse without Destroying)**

Section 2: Basic Queue Operations

★ **Enqueue N Elements and Dequeue One by One**
★ **Reverse a Queue using a Stack**
Reverse First K Elements of a Queue
★ **Check if a Queue is Empty or Full**
Sort a Queue without Extra Data Structure
Interleave First Half and Second Half of a Queue

Section 3: Sliding Window using Deque

★ **First Negative Number in Every Window of Size K**
★ **Maximum of Every Window of Size K (using Deque)**
★ **Minimum of Every Window of Size K (using Deque)**
Sliding Window Maximum (Monotonic Deque)
Count Distinct Elements in Every Window of Size K

Section 4: Queue in BFS / Level Order

★ **Level Order Traversal of Binary Tree using Queue**
★ **BFS on Graph using Queue**
Rotten Oranges Problem (Multi-Source BFS)
★ **01 Matrix — Distance to Nearest 0 (BFS)**

Section 5: Queue-Based Design Problems

★ **First Non-Repeating Character in a Stream**
★ **Implement Stack using Two Queues**
Task Scheduler using Queue
★ **Circular Queue based Hot Potato Simulation**

Generate Binary Numbers from 1 to N using Queue

Section 6: Real-World Applications of Queue

★ **CPU Process Scheduling Simulation — Round Robin using Circular Queue**

★ **Ticket Counter Simulation — Serve Customers in FIFO Order**

★ **Printer Job Queue — Manage Print Jobs with Priority**

Hospital Patient Queue — Emergency vs Normal Queue using Priority Queue

★ **Bank Teller Simulation — Multi-Counter Queue Management**

Breadth-First Web Crawler — Crawl Web Pages Level by Level using Queue

★ **Call Center Queue — Assign Agents to Waiting Customers**

Sliding Window Network Traffic Analyser — Detect Burst using Deque